

Bioprinter Droplet Size and Reliability Test Plan

Scope and evidence

This plan is intended to identify a **robust operating window** for the acoustic bioprinter: settings that repeatedly produce one droplet of the required size, not merely a single setting that works once.

The master-thesis PDF was not present in the workspace when this plan was written. The plan is therefore based on:

- the thesis-linked acoustic droplet ejection literature in `Literature Main/` ;
- the archived printer programs in `drive data/` ;
- the OPBOX calibration programs in `Dimitris Starting Files/op-box documentation/` ;
- the June 2024 experimental images, notes, and generated QA results.

No OPBOX GUI or original printer program needs to be modified to execute this plan. Test settings can be changed in copied NC/G-code test programs, and the results can be analyzed offline using the existing local Python tools.

Main findings

Factors expected to affect droplet size

Priority	Factor	Expected effect
1	Ultrasound frequency	Higher frequency generally produces smaller droplets. Frequency also affects the usable power and focus window.
1	Ultrasound amplitude / acoustic power	Increasing power helps cross the ejection threshold and generally increases droplet velocity and volume. Excessive power can cause double droplets, satellites, or splashing.
1	Pulse duration / cycle count	Longer pulses or more cycles generally increase droplet volume. Too few cycles cause missing droplets; too many may cause doubles or satellites.
1	Focus position, Z/depth, OPBOX peak alignment	Small focus errors can change ejection from complete failure to 36/36 droplets. The robust depth range matters more than the single best depth.
2	Liquid viscosity, surface tension, and density	Higher viscosity generally requires more acoustic power. Surface tension and viscosity affect ejection threshold, velocity, break-off, and final droplet size.
2	Temperature	Changes fluid viscosity and surface tension and may also change transducer/coupling behavior.
2	Liquid level / reservoir fill height	Changes the distance between the acoustic focus and the free liquid surface. Evaporation and depletion can therefore create drift during a run.
3	Trigger delay and repetition timing	Can change the interaction between signals and whether the liquid surface settles before the next ejection.
3	Target-substrate distance and surface properties	Affect spreading after impact, so measured deposited diameter may change even when in-flight droplet volume is unchanged.

The literature specifically reports that droplet size is inversely related to frequency, and that amplitude, excitation duration, fluid properties, and focus affect droplet size or velocity. It also reports that pulse width and peak pressure can increase droplet volume.

Factors expected to affect reliability

Priority	Factor	Failure mode to watch
1	Gas bubbles and insufficient degassing	Missing droplets, unstable OPBOX peak, temporary double droplets, or loss of ejection
1	Focus and OPBOX peak drift	Abrupt transition between no ejection, partial grids, and reliable ejection
1	Acoustic amplitude and cycle count	Missing droplets below threshold; doubles, satellites, and splashing above the useful range
1	Temperature and cooling-loop stability	Gradual size drift, peak drift, and changing ejection success
1	Liquid-level drift, evaporation, and material age	Time-dependent focus and droplet-size drift
2	Dwell/settling time between movement and ejection	Poor placement, doubles, or unstable ejection caused by vibration or an unsettled liquid surface
2	Stage backlash, acceleration, calibration, and substrate movement	Grid skew, unequal X/Y pitch, poor placement, or local rows/columns shifted
2	Coupling, transducer alignment, and contamination	Reduced acoustic transfer and day-to-day variation
3	Camera, lighting, and image-analysis thresholds	Apparent failures that are actually detection errors

The June 19 note is especially important: after more than one hour of degassing and purging bubbles from the cooling loop, accuracy was immediately good, but double droplets appeared from the third grid and then stopped after about three grids. This shows that preparation and **time/order effects** must be tested explicitly.

The existing June 20 results also show a sharp focus-related transition. At fixed filename-encoded settings of BP 42.93 , 100 mV , and Cycle 190 , several nearby positions produced between 0 and 33 droplets, while positions 39.13 through 39.20 produced 36/36. Later "After test" images remained uniform within each grid but their mean radius ranged from approximately 17.31 px to 19.59 px, a roughly 13% change. Reliable ejection alone therefore does not guarantee stable droplet size.

Response variables

Record the following for every grid:

1. **Ejection success:** counted droplets / 36 attempted droplets.
2. **Confirmed double-droplet rate:** manually confirmed doubles / attempts.
3. **Confirmed satellite rate:** manually confirmed satellites / attempts.
4. **Mean deposited diameter and diameter/radius CV.**
5. **Diameter drift:** percentage change from the first control grid.
6. **Placement quality:** grid-quality score, placement RMS, spacing CV, and X/Y pitch imbalance.
7. **Process drift:** OPBOX peak, aligned Z/depth, liquid temperature, reservoir level or mass, and elapsed run time.
8. **Time to failure:** grid number and droplet number at the first miss, double, satellite, or splash.

The double/satellite detector should be treated as a screening tool. Manually review all flagged droplets during the first experiments to estimate its false-positive rate.

Initial acceptance criteria

These are starting criteria. Freeze the final limits only after completing the three-day baseline.

Measure	Screening pass	Qualification target
Ejection success	At least 35/36 per grid	At least 99% across the qualification run
Confirmed doubles	No more than 1 per 108 attempts	No more than 0.5%
Confirmed satellites	No more than 2 per 108 attempts	No more than 1%
Radius or diameter CV within a grid	At most 10%	At most 7%
Mean diameter drift within a block	At most 8%	At most 5%
Grid-quality score	At least 75	At least 85
Placement RMS	At most 15% of pitch	At most 10% of pitch

Use a stricter biological acceptance criterion if the target application requires a specific deposited volume, cell count per droplet, or cell viability.

Controls and logging

Keep these fixed unless they are the factor being tested:

- one fluid/material batch per experimental block;
- material concentration, preparation method, and age;
- reservoir, starting fill volume, substrate, target distance, and camera;
- image scale, lighting, focus, and analysis settings;
- current qualified ultrasound frequency;
- grid definition: 6 x 6 , 36 attempted droplets;
- stage speed, grid pitch, and print direction;
- room conditions as far as practical.

Log these fields for every grid:

```
run_id, date_time, operator, day, material_lot, material_age_min,
material_concentration, viscosity_if_known, surface_tension_if_known,
degas_time_min, cooling_loop_purged, visible_bubbles,
reservoir_start_mass_g, reservoir_current_mass_g, liquid_level_mm,
liquid_temp_start_C, liquid_temp_end_C, room_temp_C,
US_frequency, US_amplitude, US_cycle_count, CH2_frequency, trigger_delay,
OPBOX_target_peak, OPBOX_peak_before, OPBOX_peak_after,
start_Z_depth, aligned_Z_depth, fine_step, dwell_time_s,
grid_number, image_name, notes
```

At the start and end of every test block, print the same control condition. This detects drift that would otherwise be mistaken for a factor effect. Randomize the order of test settings within a block, except during deliberate long-run/time-drift tests.

Concrete staged test plan

Stage 0: Measurement-system and baseline check

Purpose: confirm that the camera and analysis can distinguish real printer variation from measurement variation.

1. Prepare the printer using the current standard procedure.
2. Use the current best-known setting. The archived images suggest starting from 100 mV , Cycle 190 , and the current 5 MHz-class setup, but confirm the actual controller values before testing.
3. Print three consecutive 6 x 6 grids without changing any setting.
4. Photograph each grid three times without moving it.
5. Repeat on three independent days.

Minimum workload: 27 analyzed images and 324 attempted physical droplets.

Decision gate: repeated photographs of the same grid should differ by less than 2% in mean diameter and less than 2 percentage points in droplet count. Fix lighting, scale, focus, or analysis thresholds before continuing if this gate fails.

Stage 1: Preparation, bubbles, and transient reliability

Purpose: determine whether degassing, cooling-loop bubbles, and warm-up explain early-run doubles or failures.

Test the following six conditions:

Factor	Levels
Degassing time	0, 30, and 60 minutes
Cooling loop	deliberately not fully purged, and fully purged/no visible bubbles

For each condition:

1. Record starting temperature, fill level/mass, OPBOX peak, and aligned depth.
2. Print ten consecutive grids at the fixed baseline setting.
3. Record temperature, peak, and fill level after grids 1, 3, 5, and 10.
4. Photograph and analyze every grid.
5. Repeat the six-condition block on three days, randomizing condition order.

Workload: 180 grids and 6,480 attempted droplets.

Decision gate: select the preparation method that minimizes misses, confirmed doubles, satellites, and peak/diameter drift. If any bubbles are associated with large failures, require a fully purged loop for all later tests.

Stage 2: Focus, depth, and OPBOX alignment window

Purpose: locate a broad reliable focus plateau and quantify alignment repeatability.

1. Use the best preparation method from Stage 1.
2. Hold frequency, amplitude, cycles, trigger delay, dwell, and fluid constant.
3. Center the sweep on the current qualified aligned depth.
4. Test nine offsets: -0.40 , -0.30 , -0.20 , -0.10 , 0.00 , $+0.10$, $+0.20$, $+0.30$, and $+0.40$ mm.
5. At every offset, print three grids.
6. Print the center control after every third offset.
7. Repeat on two days.

Also compare:

- fixed Z/depth versus OPBOX peak-aligned depth;
- current OPBOX fine step versus one smaller fine step;
- current dwell versus a longer alignment dwell.

Core sweep workload: 54 grids and 1,944 attempted droplets, plus controls.

Decision gate: choose the center of the widest consecutive range that passes the screening criteria, not the point with the single highest score. The selected operating depth should have at least 0.20 mm margin on both sides if the hardware permits it.

Stage 3: Acoustic waveform window

Purpose: determine the amplitude/cycle combinations that eject one stable droplet and control size.

Use a 3 x 3 amplitude/cycle screen around the current nominal values:

Factor	Low	Center	High
Amplitude	nominal -15%	nominal	nominal +15%
Cycle count	nominal -15%	nominal	nominal +15%

For the archived 100 mV , Cycle 190 condition, this corresponds approximately to:

- amplitude: 85 , 100 , 115 mV ;
- cycles: 160 , 190 , 220 .

Confirm that these levels are within the equipment and material limits before running them.

1. Randomize the nine conditions.
2. Print three grids per condition.
3. Print a center control at the start, middle, and end.
4. Repeat the complete block on two days.
5. After selecting the best amplitude/cycle region, test trigger delay at five levels centered on the current value. Use the existing controller's safe delay range and step size.

Core amplitude/cycle workload: 54 grids and 1,944 attempted droplets, plus controls.

Decision gate: reject settings below the reliable ejection threshold and settings that increase doubles, satellites, or splashing. Among the remaining settings, choose the lowest-energy region that achieves the required size and passes reliability limits.

Treat ultrasound frequency as a separate experiment. Frequency is a primary droplet-size control but changing it may require different hardware, calibration, and power levels. Only test alternative frequencies after the nominal-frequency operating window is stable.

Stage 4: Fluid, temperature, and liquid-level robustness

Purpose: measure how sensitive the qualified waveform is to realistic material and environmental variation.

Run one factor at a time first:

Factor	Suggested levels
Liquid temperature	target -2 C, target, target +2 C
Starting fill level or mass	low, nominal, high within the normal operating range
Material age after preparation	fresh, mid-run, maximum intended use time
Material concentration / viscosity	low, nominal, high specification limit

For each factor, print three grids per level plus start/end controls, and repeat on three days or three independent material batches.

Decision gate: if a factor changes mean diameter by more than 5% or causes a reliability failure, either tighten its operating specification or create a separate qualified waveform/focus setting for that condition.

Stage 5: Motion, dwell, and placement

Purpose: separate acoustic ejection reliability from mechanical placement reliability.

Test:

- dwell after stage movement: current, 2 x current , and a shorter value;
- slow, nominal, and fast stage speed/acceleration;
- alternating versus one-direction grid paths;
- grids printed in several bed/reservoir locations.

Print three grids per condition and repeat the block twice. Use the same qualified ejection setting throughout.

Decision gate: select the shortest dwell and fastest motion setting that still meets the placement and ejection qualification targets. Investigate backlash, vibration, substrate movement, or X/Y scale separately if placement fails while count and droplet size remain stable.

Stage 6: Final qualification and long-run test

Purpose: verify repeatability after all settings and preparation rules are fixed.

1. Freeze the preparation SOP, fluid specification, focus method, waveform, dwell, motion settings, target distance, and analysis settings.
2. On each of three days or three independent material batches, print 100 consecutive grids.
3. Analyze every grid for count, size, doubles/satellites, and placement.
4. Log temperature, OPBOX peak, aligned depth, and reservoir level/mass every ten grids.
5. Perform one stop/restart cycle after grid 50 to test recovery.

Qualification workload: 300 grids and 10,800 attempted droplets.

Pass only if every day meets the qualification targets and no systematic time trend exceeds the control limits.

Analysis and decision method

1. Plot every response against grid number/time before comparing settings. This exposes warm-up, depletion, and drift.
2. For Stage 2, plot success rate and mean diameter against depth/peak offset.
3. For Stage 3, plot amplitude-by-cycle response surfaces for success, size, doubles, and satellites.
4. Analyze ejection success as a binary response and diameter as a continuous response. Include day/material batch as a blocking variable.
5. Prefer settings in the middle of a passing region. Avoid settings directly next to no-ejection or double-droplet regions.
6. Manually review a random 10% of unflagged images and all flagged images until the automated detector's error rate is known.

Useful outputs already produced by the local analysis workflow are:

- `droplet_count_summary.csv` ;
- `droplet_size_uniformity_summary.csv` ;
- `droplet_double_satellite_summary.csv` ;
- `droplet_grid_quality_summary.csv` .

Stop rules

Stop a condition immediately and record it as a failure if:

- there is visible splashing or unsafe transducer behavior;
- no droplets eject for two consecutive rows;
- confirmed doubles exceed 10% in a grid;
- a bubble enters the acoustic path;
- temperature leaves the approved range;
- OPBOX peak/alignment cannot converge;
- the substrate or reservoir visibly moves.

Do not silently adjust a setting during a test block. Record the failure, return to the control condition, and start a new labeled block after any intervention.

Recommended first experiment

Start with **Stage 1**, because the local note already identifies degassing, cooling-loop bubbles, and transient double droplets as likely reliability drivers. Use the current nominal waveform and the current

best OPBOX-aligned depth. Once the preparation procedure is stable, run the Stage 2 focus sweep; otherwise waveform optimization will be confounded by bubbles and drift.